Haotian Zhang

hzhang44@usc.edu

Education

University of Southern California Starting Spring 2027
Incoming undergraduate in Applied and Computational Mathematics.

American University of Paris August - December 2026
Visiting Student · USC Fall Semester in Paris

Research Interests

Video Reasoning; World Models; Foundation Models; Applied Mathematics.

I currently study how video models represent the physical world and reason about changes over time. During the study, I design cognitively grounded tasks to train video models and evaluate their reasoning abilities.

Honors and Awards

Hugging Face #1 Paper of the Week September 2026
For *Training Object Permanence in World Models*.

United States of America Mathematical Olympiad (USAMO) 2024
Qualifier; American Invitational Mathematics Examination (AIME): 13/15.

American Mathematics Competition (AMC 12) 2024
Global Top 1%.

Berkeley Math Tournament 2023
National High Distinction (individual); national Top 2 overall and Top 3 in the Guts round.

S.-T. Yau High School Science Award - Mathematics 2024
Second Prize (Top 8), Southern Region Finals.

Publications

* Equal contribution. Ellipses shorten long author lists without changing the original order.

Preprint

Training Object Permanence in World Models

Haotian Zhang*, Fengyuan Yu*, Dezhi Luo*, Haoran Sun, Zehong Zhao, Qingying Gao, ..., Yilun Du, Hokin Deng.

arXiv:2609.28654, 2026. Co-first author.

Paper | Project | Code

Research Experience

Hippocampal June 2026 - Present
Research Intern

Training Object Permanence in World Models (WROP) June - September 2026
Co-first author, core team member

- Designed 150 Blender-based 3D task generators to study object permanence and solidity, supporting the project's 1.5-million-sample training corpus.
- Contributed to training a 16B-parameter video model, human evaluation, and manuscript preparation.
- Co-first authored the resulting preprint, recognized as Hugging Face's #1 Paper of the Week.

Research Experience (continued)

University of California, Santa Cruz June - August 2025
Research Intern, Science Internship Program (SIP)

- Developed a generative-AI tool to scaffold research-paper reading through simulated reader-AI dialogue and iterative refinement of system prompts.
- Designed and evaluated prompts to help AI systems explain complex academic texts.
- Evaluated NotebookLM's Audio Overview for vocal tone, pacing, and content accuracy against professional podcast standards; explored prompt strategies for more natural narration.

Cambridge Centre for International Research (CCIR) June - September 2025
Research Student

- Completed coursework in Applied Linguistics & Big Data.
- Authored a research paper on the social functions of multilingualism along the ninth- and tenth-century Silk Road, combining linguistic theory with data-informed analysis.